

# Modeling Transfer Miss Probability in NYC Subway Systems

A Stochastic Approach with Crowd Density Effects

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# Motivation: The Transfer Problem

- NYC commuters (like myself) frequently miss connecting trains at transfers
- **Significant Uncertainties:**
  - Random passenger position on platform
  - Variable walking speeds due to crowds
  - Unpredictable train headways
- **Goal:** Quantify probability of missing transfers to help transit agencies and passengers to identify their chances of making their transfer.

## Core probability statement:

$$P_{\text{miss}} = P(\mathbb{W} \geq \mathbb{G} \mid Q_i \subset \mathcal{P}, T = t_i) \quad (1)$$

where:

- $\mathbb{W} = S_t + D_t$  : total walking time
- $S_t$  : platform  $\mathcal{P}$  walking time (dependent on position  $Z \sim \text{Uniform}(0, L)$ )
- $D_t$  : inter-platform distance (Haversine formula)
- $\mathbb{G} : N_t - F_t =$  headway gap time between trains
- $T = t_i$  : time of the day on a 24 hour scale

# Mathematical Formulation

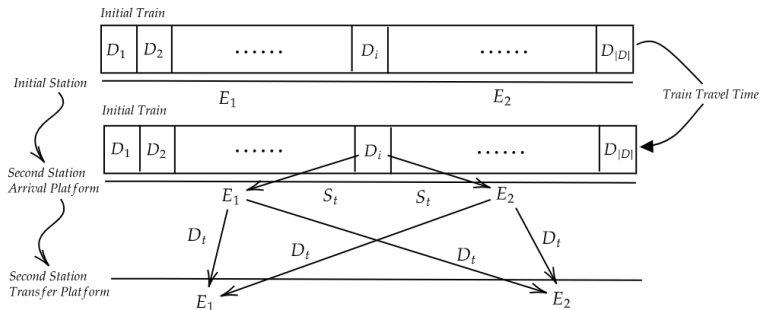


Figure 1: One step Transfer Process

**Key innovation:** Walking speed varies with crowd density so we can express  $W$  as

$$W = \frac{2R \arcsin \left( \sqrt{\sin^2 \left( \frac{\Delta\phi}{2} \right) + \cos(\phi_i) \cos(\phi_j) \sin^2 \left( \frac{\Delta\lambda}{2} \right)} \right) + Z - \mathcal{L}(E_\alpha)}{v_0 \cdot \left( 1 - \frac{\beta}{1 + e^{-k(\rho t - \rho c)}} \right)}$$

# Model Components

**Case study:** Grand Central–42nd St  
(4 train Woodlawn Bound  $\rightarrow$  7 train Flushing Bound)

## Spatial variables:

- Platform length  $L = 155.5\text{m}$
- Stair exit location  $E_\alpha$  on 4/5/6 platform in Grand Central
- Haversine distance: 78.078m

## Temporal variables:

- Crowd density  $\rho_t$  (bimodal Gaussian)
- Base speed  $v_0 = 1.4 \text{ m/s}$
- Four time periods (Late Night, Morning Rush, Afternoon, Evening)
- Simulated data for  $\mathbb{G}$
- Spatial data from DATA.NY.GOV

# Solution Approach

- ① Simulated  $N = 5000$  passenger transfers
- ② Incorporated realistic parameters:
  - Time-varying crowd density  $\rho_t$  (peaks at 8:30 AM, 5:30 PM)
  - Sigmoid walking speed function
  - Actual headway distributions by time of day
- ③ Filtered impossible transfers (21.8%)
- ④ Analyzed  $n = 3908$  realistic transfer attempts after the clip function
- ⑤ Validated using ROC curve analysis

# Results and Model Performance

## Key findings:

- Overall miss rate: **34.57%**
- Mean walking time: 130.2s
- Mean headway gap: 133.5s
- Critical margin: 3.3s

## Model validation:

- AUC = **0.9391**
- Accuracy: 86%
- Conservative bias

|                | Predicted Success | Predicted Miss |
|----------------|-------------------|----------------|
| Actual Success | 2256              | 301            |
| Actual Miss    | 243               | 1108           |

**Insight:** Small differences between  $\mathbb{W}$  and  $\mathbb{G}$  dramatically impact transfer success



# Conclusions and Future Work

## Achievements:

- Successfully quantified transfer risk using stochastic framework
- Integrated crowd dynamics, platform geometry, and headway variability
- Strong predictive performance for real-world application

## Limitations & Extensions:

- Haversine distance underestimates actual walking paths
- Integrating real-time MTA feed for dynamic predictions
- Using Bayesian prior distributions from observed passenger behavior

**Impact:** Provides quantitative tool for transit agencies to identify high-risk transfers and develop real-time passenger guidance systems

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